

# Molecular Simulations of CO<sub>2</sub> and H<sub>2</sub> Sorption into Ionic Liquid 1-*n*-Hexyl-3-methylimidazolium Bis(trifluoromethylsulfonyl)amide ([hmim][Tf<sub>2</sub>N]) Confined in Carbon Nanotubes

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Atomistic simulations are used to study the ionic liquid (IL) 1-*n*-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide ([hmim][Tf<sub>2</sub>N]) confined into (20,20) and (9,9) carbon nanotubes (CNTs) and the effect of confinement upon gas sorption. The cations and the anions exhibit highly ordered structures in the CNT. There are more cations adsorbed close to the (20,20) tube wall while more anions adsorb in the tube center at high IL loadings. The IL molecules in the CNT exhibit self-diffusivity coefficients about 1–2 orders of magnitude larger than the corresponding bulk IL molecules. Sorption of CO<sub>2</sub> and H<sub>2</sub> gases in the composite material consisting of CNT and IL indicates that H<sub>2</sub> molecules diffuse about 1.5 times faster than the CO<sub>2</sub>. In contrast, H<sub>2</sub> diffuses about 10 times faster than CO<sub>2</sub> in both the CNT and in bulk IL. The CNT exhibits the largest amount of sorption for both CO<sub>2</sub> and H<sub>2</sub>, followed by the composite material, and the IL exhibits the least gas sorption. When the temperature is increased, the amount of sorbed CO<sub>2</sub> decreases in all three types of systems (IL, CNT, and the composite material) while the H<sub>2</sub> sorption increases in [hmim][Tf<sub>2</sub>N], decreases in the CNT, and does not change significantly in the composite material. The composite material exhibits higher sorption selectivity for CO<sub>2</sub>/H<sub>2</sub> than both the IL and the CNT. It is very interesting to note that the IL molecules can be dissolved in the CO<sub>2</sub> molecules under confinement due to a favorable negative transferring energy. However, in the absence of confinement the IL molecules will not dissolve in the CO<sub>2</sub> due to a very large unfavorable positive transferring energy.

## 1. Introduction

Room-temperature ionic liquids (RTILs) have attracted significant attention both from the industry and from the academia due to their special properties.<sup>1–3</sup> In particular, the nonvolatile properties of RTILs are highly desired in chemical process.<sup>4</sup> The properties of the ionic liquids can be tuned either through coupling different cations and anions or by covalent tethering task-specific functionalities<sup>5,6</sup> to one or both of the ions.

Experimentally, it has been found that ionic liquids confined in nanospaces exhibit many novel properties which are different from those of the bulk ionic liquids. For example, either decrease<sup>7</sup> or increase<sup>8</sup> of the melting point has been observed under confinement. Very recently, it was also found that the solubility of CO<sub>2</sub> in the IL could be enhanced through confinement.<sup>9</sup>

Molecular simulations have been performed to study IL confined in nanoporous materials and adsorbed on surfaces.<sup>10–15</sup> A density enhancement for both the cation and anion species has been observed for ILs confined into the silica slit pore<sup>10</sup> or adsorbed on a graphite surface.<sup>12</sup> The interactions between ionic liquids and carbon nanotubes (CNTs) have also been investigated. Cylindrical shell-like distributions of the ionic liquid outside of the nanotubes have been observed, while the ionic liquid structures inside the nanotubes vary markedly with their diameter.<sup>14</sup> When the ionic liquid is confined into silica slit

pores, in most cases, the self-diffusion coefficients of the ionic liquid in a direction parallel to the slit pore is about two times faster than those of the bulk ionic liquid.<sup>10</sup> Free energy calculations based on Monte Carlo simulations suggest that filling the CNT by ion pairs from the bulk IL is more favorable than for single ions.<sup>15</sup>

The previous studies are important for understanding the behavior of the ionic liquid itself under either confinement or when adsorbed on surface. In this study, we expand the previous investigations and consider other important questions, namely what is the effect of the confinement of the IL on gas separation properties? Does the confined IL behave very differently from the bulk ionic liquid upon gas absorption? Does the composite material consisting of the IL and CNT present improved properties in terms of gas separation? In order to address the above questions, we used atomistic Monte Carlo (MC) and molecular dynamics (MD) simulations. As far as we know, this is the first time that gas sorption properties of an ionic liquid under nano confinement have been considered based on atomistic simulations.

In this study, we consider a CNT as the confinement material. This selection presents the advantage of an easier system to model due to the 1-dimensional (1-D) structure involved. However, we expect that the novel behavior of the 1-D confined ionic liquid for gas separation might set the stage and be also useful in future studies of confinement in more complex and practical structures. In spite of the simplicity of our model consisting of a single-wall CNT, some of the predicted properties from this study, such as the thermodynamic properties, might also be tested experimentally for ILs confined in multiwall

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carbon nanotubes. This is due to the fact that we are using a large CNT with a diameter of about 27 Å, which is close to the experimental pore size for multiwall carbon nanotubes with diameters of 31–75 Å.<sup>16</sup>

## 2. Theory

**2.1. NP<sub>zz</sub>T Ensemble.** We have used the NP<sub>zz</sub>T statistical ensemble (constant number of particles, constant *z*-component pressure tensor, and constant temperature) to compute the thermodynamic properties of the pure ionic liquid confined into 1-D CNT. In this ensemble, the fundamental thermodynamic equation describing a system of *N* adsorbate particles inside the CNT is given by

$$dG = -S dT + \mu dN + T_R 2\pi R H_z dR - H_z d(T_{zz} \pi R^2) \quad (1)$$

where  $G = \mu N$  is the Gibbs free energy,  $\mu$  is the chemical potential for the adsorbate (the adsorbed ionic liquid),  $T$  is the temperature,  $S$  is the entropy,  $R$  is the radius of the carbon nanotube defined as the distance from the tube center to the carbon atom center on the tube wall, and  $H_z$  is the length of the tube. As pointed out by Panagiotopoulos,<sup>17</sup> the choice of  $R$  will not change the thermodynamic properties obtained from Monte Carlo simulations as long as the regions with essentially nonzero density at equilibrium are included in  $R$ . In eq 1  $T_R$  and  $T_{zz}$  represent the radial and axial stresses along respective axes of the nanotube, and they are equal to the negative values of the corresponding pressure tensor components. By specifying the values of  $T$ ,  $N$ ,  $T_{zz} = -P_{zz}$ , and  $R$  variables, we performed NP<sub>zz</sub>T MC simulations to calculate the average molar volume of the ionic liquid confined in a specified CNT. The acceptance rules for NP<sub>zz</sub>T ensemble are very similar to those used in generic NPT MC simulations, except for the change of the tube length in the  $Z$  direction, rather than the volume change as in the NPT ensemble. Specifically, for the  $Z$ -length type of move we used the following procedure:

1.  $H_{z,\text{new}} = H_{z,\text{old}} + \Delta H_{z,\text{max}} \times (2.0 \times \xi - 1)$ , where  $\Delta H_{z,\text{max}}$  is the maximum length change in  $Z$  direction,  $\xi$  is a random number between 0 and 1,  $H_{z,\text{new}}$  and  $H_{z,\text{old}}$  are the tube length after and before the  $Z$ -length move.

2. The  $Z$  coordinate of the first atom of each molecule is scaled to a new position by  $Z_{1,\text{new}} = Z_{1,\text{old}} \times (H_{z,\text{new}}/H_{z,\text{old}})$ , where  $Z_{1,\text{old}}$  and  $Z_{1,\text{new}}$  are the  $Z$  coordinate of the first atom of each molecule before and after the  $Z$ -length move. The other atoms of each molecule are shifted in the  $Z$  direction relative to the first atom according to relation.  $Z_{i,\text{new}} = Z_{i,\text{old}} + Z_{1,\text{new}} - Z_{1,\text{old}}$ ,  $i \geq 2$ .

3. The above-described  $Z$ -length move is accepted using the acceptance probability

$$\frac{\text{acc}(o \rightarrow n)}{\text{acc}(n \rightarrow o)} = \exp \left\{ -\beta \left[ P_{zz} \pi R^2 (H_{z,\text{new}} - H_{z,\text{old}}) + (U_{\text{new}}^{\text{inter}} - U_{\text{old}}^{\text{inter}}) - N k_B T \ln \left( \frac{H_{z,\text{new}}}{H_{z,\text{old}}} \right) \right] \right\} \quad (2)$$

where  $\beta$  is the reduced temperature,  $k_B$  is the Boltzmann constant, and  $U_{\text{new}}^{\text{inter}}$  and  $U_{\text{old}}^{\text{inter}}$  are respectively the interaction energies after and before the  $Z$ -length change. Here, the interaction energy  $U^{\text{inter}}$  consists of the adsorbate–solid interaction and the interaction between the atoms of different adsorbate molecules as well as the interaction between the atoms of the same adsorbate molecule when these atoms are separated by at

least three consecutive bonds. Examples of such intramolecular interactions considered here are the 1–4, 1–5, 1–6, etc., interactions. In eq 2 the interaction energies are obtained as a summation over the total number of cation and anion species of the system. Note that, when the adsorbate–adsorbate interaction is negligibly small, one can obtain the ideal gas law equation, i.e.,  $P_{zz} \pi R^2 H_z = N k_B T$  in the 1-D confinement. For a simple Lennard-Jones fluid, when the pressure tensor  $P_{zz}$  is small, the interaction energy between the adsorbate molecules is negligibly small. This corresponds to a very low adsorbate loading. In this case, the ideal gas law was found to hold in our simulations as expected. Additionally, the computed pressure tensor axial component was found to be equal to the imposed one in the simulation which confirms that we have correctly implemented the pressure tensor calculations in our in-house code.

**2.2. Isostress–Osmotic Ensemble.** We have used isostress–osmotic ensemble to calculate gas sorption in ionic liquid confined into CNT. The isostress–osmotic ensemble is a combination of the grand-isostress<sup>18</sup> and the osmotic ensemble.<sup>19</sup> The fundamental thermodynamic equation for the isostress–osmotic ensemble is given by

$$dG_2 = -S dT - N_1 d\mu_1 + T_R 2\pi R H_z dR - H_z d(T_{zz} \pi R^2) + \mu_2 dN_2 \quad (3)$$

where  $G_2 = \mu_2 N_2$ ; 1 and 2 denote the solute and the solvent molecules, respectively. The meaning of other symbols in eq 3 is the same as in eq 1. By specifying  $T$ ,  $\mu_1$  ( $f_1$ ),  $T_{zz}$  ( $-P_{zz}$ ),  $N_2$ , and  $R$ , where  $f_1$  is the fugacity for the solute gas, we perform isostress–osmotic Monte Carlo simulations. Note that simulations performed in isostress–osmotic ensemble are similar to those in the osmotic ensemble, in which  $T$ ,  $f_1$ ,  $P$ , and  $N_2$  variables are specified.<sup>20</sup> The only difference is that the pressure  $P$  in the osmotic ensemble is replaced by the pressure tensor axial component  $P_{zz}$  in the isostress–osmotic ensemble. Additionally, in the isostress–osmotic ensemble, it is also necessary to specify the radius  $R$  of the CNT.

We used the fractional component method<sup>20</sup> to insert and delete solute molecules in the isostress–osmotic ensemble. The acceptance rules in the continuous fractional component (CFC) isostress–osmotic Monte Carlo (MC) ensemble are very similar to those for the CFC osmotic ensemble as described in our previous work.<sup>20</sup> Specifically, in the isostress–osmotic MC simulations there are three types of moves, i.e., the thermal move, the  $H_z$  move, and the  $\lambda$  move, where  $\lambda$  is the coupling strength between the fractional molecule and the other solute and solvent molecules. Additional details about this method can be found in our previous work.<sup>20–22</sup> We note that no scaling of the interaction between the fractional solute molecule and the CNT was performed in the current case. Instead, when the new fractional gas molecule is inserted in a position separated by less than 1.0 Å from the nanotube wall, the attempted insertion is simply rejected.

The thermal move in the isostress–osmotic ensemble is exactly the same as in the osmotic<sup>20</sup> and NP<sub>zz</sub>T ensembles. The  $H_z$  move in the isostress–osmotic ensemble is the same as in the NP<sub>zz</sub>T ensemble discussed above excepting the acceptance rule. Specifically, the number of particles  $N$  in eq 2 for the NP<sub>zz</sub>T ensemble is replaced by  $N_1 + 1 + N_2$ , where  $N_1$  is the number of full solute molecules, the number 1 (not the subscript) corresponds to one fractional solute molecule and  $N_2$  is the total number of cations and anions. The  $\lambda$  move to insert and delete

molecules in the CFC isostress–osmotic ensemble is the same as in the CFC osmotic ensemble,<sup>20</sup> except that the volume  $V$  in the osmotic ensemble (see eqs 14 and 15) in the previous work<sup>20</sup> is replaced by  $2R \times 2R \times H_z$  in the isostress–osmotic ensemble. In the current case the new fractional molecule is attempted to be inserted in a box with sides of  $2R \times 2R \times H_z$  which contains the carbon nanotube. During MC moves, when the inserted molecules are outside the carbon nanotube, we simply reject the insertion move and the old configuration is collected for statistics. One can also use larger boxes, for example with dimensions of  $2.5R \times 2.5R \times H_z$ . Our simulations show, however, that the results are almost the same using these two boxes with different sides and, consequently, in this work we have used consistently only the smaller box with sides of  $2R \times 2R \times H_z$ .

### 3. Simulation Details

**3.1. Classical Force Field.** A classical force field has been used to simulate the ionic liquid [hmim][Tf<sub>2</sub>N], the CNT and their interactions with sorbed CO<sub>2</sub> and H<sub>2</sub> molecules. The interaction energy of the system is given by

$$\mathcal{V}(\mathbf{r}) = \sum_{\text{bonds}} k_b(r - r_0)^2 + \sum_{\text{angles}} k_\theta(\theta - \theta_0)^2 + \sum_{\text{dihedrals}} k_\chi[1 + \cos(n_\chi\chi - \delta_0)] + \sum_{\text{impropers}} k_\psi(\psi - \psi_0)^2 + \sum_{i=1}^{N-1} \sum_{j=i+1}^N \left\{ 4\epsilon_{ij} \left[ \left( \frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left( \frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{r_{ij}} \right\} + U_{\text{sf}} \quad (4)$$

where the symbols have their conventional meaning,<sup>23</sup>  $U_{\text{sf}}$  is the interaction between the adsorbates (the ionic liquid and the gas solute molecules) and the CNT. For the CFC MC simulations, the Lennard-Jones (LJ) parameters for the fractional molecule were scaled using the functional form proposed by van Gunsteren et al.<sup>24</sup> to aid insertion when computing the interaction between the fractional molecule and other adsorbate molecules. The LJ potential was switched at 10.5 Å and truncated at 12.0 Å. A Verlet neighbor list with a 13.5 Å radius was used. The intramolecular electrostatic and LJ interactions for atoms separated by exactly three consecutive bonds were scaled by 0.5 and were neglected for atoms separated by less than three consecutive bonds. The parameters for the particle mesh Ewald (PME) depend on the length of the ionic liquid confined in the carbon nanotube. For ionic liquid confined in the (20,20) carbon nanotube, the PME grid size in NAMD simulations was typically set to between 90 and 108. Force field parameters for [hmim][Tf<sub>2</sub>N] and CO<sub>2</sub> were taken from our previous work.<sup>22,25</sup> The LJ potential parameters for the C atoms of the CNT were set to be  $\sigma = 3.4$  Å and  $\epsilon/k_B = 28$  K.<sup>26</sup> The carbon nanotube was modeled to be rigid. A two-center LJ Cracknell potential<sup>27</sup> was used for H<sub>2</sub> molecule with parameters  $\sigma = 2.59$  Å and  $\epsilon/k_B = 12.5$  K. The nominal bond length  $r_0$  between the two hydrogen atoms was set to be 0.74 Å and the force constant  $k_b$  was set to be 374 kcal/(mol Å<sup>2</sup>). As we have shown before,<sup>25</sup> H<sub>2</sub> solubilities in [hmim][Tf<sub>2</sub>N] obtained using different H<sub>2</sub> potential models, i.e., the Cracknell<sup>27</sup> and the Buch<sup>28</sup> models, are close to each other. We assume that this finding continues to be valid for the case of H<sub>2</sub> sorption in the composite material and consequently only the Cracknell potential was used in current simulations.

The  $U_{\text{sf}}$  has been calculated in two different ways. In molecular dynamics (MD) simulations using the NAMD pro-

gram,<sup>29</sup> we calculated the interaction between the adsorbates and CNT by explicitly summing all the LJ interactions between each atom of the adsorbate molecules and each carbon atom of the CNT. We used the atom–atom interaction between adsorbate atoms and the carbon atoms on the nanotube because the atomistic details on the carbon nanotube have to be considered explicitly in MD simulations to get reliable self-diffusivity data.

An alternative description of the interaction potential between the adsorbates and CNT can be obtained by integrating out the angular and the Z-dependence (height) of the interactions in a unit cell. The resulted potential is only a function of the distance  $r$  from the tube center. More details about this approach are available from previous work.<sup>30</sup>

If one uses this integral model, there is no force in the Z direction. One might conclude that as a result of this integral approach the center of mass for the entire system of all adsorbate molecules will not move in the Z direction if the initial velocity of the center of the mass of the adsorbates is set to be zero. However, this is not true in the case of the atomistic potential describing the CNT.<sup>31</sup> When the atomistic potential between the adsorbate molecules and the carbon atoms on the nanotube is used, there will be a nonzero instantaneous force in the Z direction even though the average force in the Z direction on the entire adsorbate molecules due to the carbon nanotube is zero. It is this nonzero instantaneous force in the Z direction that drives the center of mass for the entire adsorbates system to drift and our simulations for CH<sub>4</sub> adsorption in carbon nanotubes (not presented here) have shown this. In our MD simulations we have used the atomistic potential instead of the integral model to compute the interactions between adsorbates and the CNT to obtain the self-diffusivity values.

In MC simulations, the  $U_{\text{sf}}$  was calculated using the integral model. We have pretabulated the interaction energy between all types of atoms and the CNT. The grid distance in the  $r$  tabulation was set to be 0.2 Å. This interaction table was loaded from the start of the simulation and it was used to calculate the energy between each atom of the adsorbate molecule and the CNT by interpolation during the simulation. The force on the adsorbate atoms due to the CNT was estimated from the preloaded interaction table also through interpolation. As described above, the Z-directional energy dependency has been averaged out through integration, hence, there is no force in Z direction on the adsorbates due to the CNT in the case of integral potential. Note that in MC simulations, it is the energy not the force that determines the acceptance of the move, and the instantaneous nonzero force in the Z direction from the atomistic  $U_{\text{sf}}$  model will not affect the simulated thermodynamic properties using the integral potential.

We have used both the Ewald summation<sup>23</sup> and the Fennel and Gezelter shift force (FGSF)<sup>32</sup> method to calculate the electrostatic interactions in MC simulations. In the case of FGSF method, a cutoff value of 12.0 Å was used with a damping parameter  $\kappa = 0.2022$  Å<sup>-1</sup>. For the Ewald method, the cutoff distance and the  $\kappa$  values were set to be 12.0 Å and 0.3334 Å<sup>-1</sup> for CO<sub>2</sub> adsorption in the (20,20) CNT with a length of 30 Å, 42 Å and 0.095 238 Å<sup>-1</sup> for an IL confined in the (20,20) CNT with a length of about 90 Å, 293 Å and 0.013 652 Å<sup>-1</sup> for an IL confined in the (9,9) CNT with a length of about 590 Å. As shown in the Supporting Information, the Ewald summation and the FGSF methods give thermodynamic properties very close to each other. In this paper, in MC simulations we used the FGSF method to calculate the electrostatics due to its increased speed up, about eight times faster than the Ewald summation method.<sup>22</sup> In MD simulations we used the particle-mesh Ewald



**TABLE 1: Molar Volume  $V_m$  Obtained from  $NP_{zz}T$  Simulations for [hmim][Tf<sub>2</sub>N] Confined in the (20,20) and (9,9) Carbon Nanotubes at 313 K and Different Axial Pressure Tensors<sup>a</sup>**

| tube    | $P_{zz}$ set (bar) | $P_{zz}$ sim (bar) | $V_m$ (cm <sup>3</sup> /mol) | $H_z$ (Å) | init $H_z$ (Å) |
|---------|--------------------|--------------------|------------------------------|-----------|----------------|
| (20,20) | 1                  | 11 ± 6             | 486.8 (6)                    | 84.0 (1)  | 95.92          |
| (20,20) | 1                  | 3 ± 11             | 471.9 (6)                    | 81.4 (1)  | 78.704         |
| (20,20) | 1                  | 2 ± 14             | 482.0 (8)                    | 83.1 (1)  | 84.6           |
| (20,20) | 500                | 500 ± 5            | 456.2 (4)                    | 78.7 (1)  | 90.414         |
| (9,9)   | 1                  | 1 ± 3              | 702.9 (5)                    | 598.7 (4) | 794            |
| (9,9)   | 1                  | 1 ± 2              | 703.0 (9)                    | 598.8 (8) | 586            |
| (9,9)   | 500                | 503 ± 2            | 687.0 (4)                    | 585.2 (3) | 794            |

<sup>a</sup> Simulations were started from different initial configurations and nanotube lengths  $H_z$ . Also shown are the computed pressure tensor component  $P_{zz}$ . The uncertainty in the last digit is given in parentheses.

method which is implemented in the NAMD software for accurate evaluation of the electrostatic interaction.

**3.2. Monte Carlo Simulations.** The  $NP_{zz}T$  simulations were performed at axial pressure tensors of 1 and 500 bar and at temperatures of 313, 423, and 573 K for 60 pairs of [hmim][Tf<sub>2</sub>N] confined in the (9,9) and (20,20) CNTs. A single carbon nanotube was held in the center of the simulation box with sides of 1000 Å × 1000 Å ×  $H_z$  Å. The interaction between molecules in the image cells and the primary cell is negligibly small due to the large size of the simulation box. Simulations included several million steps of equilibration followed by about 20 million steps of production runs. During the equilibration moves various MC moves were tuned to achieve roughly 50% acceptance rates. These moves were thermal equilibration via hybrid Monte Carlo<sup>33</sup> (HMC) (90%) and  $H_z$  moves (10%). The values in parentheses represent the fixed probabilities during the run. For HMC calculations we used five MD steps per MC move. The time step in MD was chosen between 0.4 and 1.2 fs. The maximum values for Z-length move were between 0.5 and 4.1 Å. These values were selected to achieve about 50% acceptance rate in the MC moves.

In the  $N_2 P_{zz} Tf_1$  simulations used to compute gas sorption into the composite material, the number of ionic liquid molecules was set to be 60 pairs and calculations in this case were performed only in the (20,20) CNT. Similar to  $NP_{zz}T$  ensemble, the isolated carbon nanotube was held to be fixed in the center of a box with large sides. For CO<sub>2</sub>, the temperature variations were set to be 333–573 K and the axial pressure tensors were between 4 and 200 bar. For H<sub>2</sub>, the temperatures and the axial pressure tensors were 313–573 K and 50–400 bar. The fugacity was calculated from the Peng–Robinson equation of state<sup>22</sup> for CO<sub>2</sub> and from molecular dynamics simulation for H<sub>2</sub> as described in our previous work.<sup>25</sup> Note that the axial pressure tensor  $P_{zz}$  in the carbon nanotube was set to be equal to the pressure in the gas phase. Equilibration runs of typically 2 million moves were carried out during which various MC moves were tuned to achieve roughly 50% acceptance rates. These moves were thermal equilibration via HMC (50%),  $\lambda$  move (45%), and  $H_z$  move (5%). Production runs of about 20 million steps were typically used. The time step was set to range from 0.4 to 1.2 fs. The maximum change of  $H_z$  was taken between 0.4 and 1.3 Å, and the CFC coupling parameter  $\lambda$  was changed uniformly up to a maximum value of 0.3–0.9. These values give acceptance rates of roughly 50% for the CO<sub>2</sub> and H<sub>2</sub> sorption. The CFC bias factor<sup>20</sup> was adjusted during the equilibration stage to achieve as closely as possible a uniform probability distribution of  $\lambda$ . Optimization of these bias factors was done using the Wang–Landau<sup>34</sup> updating scheme.

The CFC grand canonical<sup>20</sup> MC simulations were performed to compute CO<sub>2</sub> and H<sub>2</sub> adsorption into a single (20,20) CNT at the same temperatures and pressures as those used in  $N_2 P_{zz}$

$Tf_1$  simulations. Simulation details in this case were very similar to those for  $N_2 P_{zz} Tf_1$  except that there is no  $H_z$  move in the grand canonical ensemble. The tube length in the CFC grand canonical MC was set between 25 and 120 Å. The time step was 0.6–3.4 fs, and the CFC coupling parameter  $\lambda$  was changed uniformly up to a maximum value of 0.4–0.99.

**3.3. Molecular Dynamics Simulations.** Molecular dynamics simulations in the NVT and NVE ensembles were performed using the NAMD program.<sup>29</sup> Langevin dynamics with a thermostat was applied for temperature control. The thermostat damping factor was taken to be 5 ps<sup>-1</sup>, and the time step was 0.5–1.0 fs. Simulations were performed for 5–60 pairs of ionic liquids confined into the (20,20) and (9,9) CNTs at 313–573 K. Simulations were run typically for about 5 ns of equilibration followed by 10–20 ns of production runs. During the simulations, the CNT was fixed by using the *fixedAtoms* utility provided in the NAMD program. In this case, the carbon atoms of the CNT are held fixed at their original positions for the entire duration of the simulation. However, the interactions between the carbon atoms on the CNT and the adsorbate molecules are still computed. The NVT trajectories were used to analyze the ionic liquid structure confined into the carbon nanotube. The self-diffusivity in the Z direction was calculated based on the NVE MD trajectories using a procedure similar to the one described previously<sup>25,35</sup> as

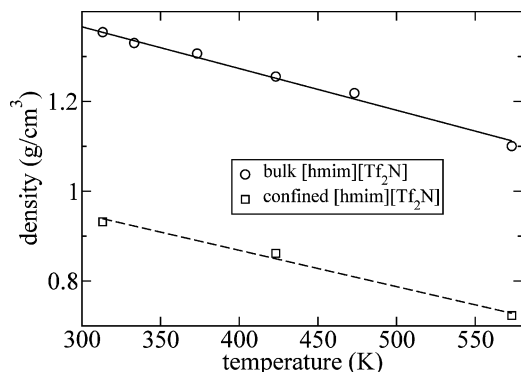
$$D_{\text{self},z} = \lim_{t \rightarrow \infty} \frac{1}{2t} \langle |z(t) - z(0)|^2 \rangle \quad (5)$$

## 4. Results

### 4.1. Thermodynamics of [hmim][Tf<sub>2</sub>N] Confined into CNT.

**4.1.1. Molar Volume and Density of [hmim][Tf<sub>2</sub>N] Confined into CNT.** The molar volumes of the ionic liquid confined in different carbon nanotubes at 313 K and for different axial pressure tensors are shown in Table 1. In the (20,20) nanotube, when the axial pressure tensor is increased from 1 to 500 bar, the molar volume of the confined ionic liquid is decreased by about 5%. In the (9,9) carbon nanotube, the molar volume of the ionic liquid is decreased by about 2% when the axial pressure tensor is increased from 1 to 500 bar. These compressibility values for the confined ionic liquid are comparable to those obtained for the bulk ionic liquid. The molar volume for the bulk [hmim][Tf<sub>2</sub>N] decreases by about 1% at 333 K when the pressure is increased from 1 to 200 bar. Note that the simulated uncertainty for the confined ionic liquid molar volume is about 3% as shown below and hence no clear trends in compressibility values with tube diameter can be established.

We have also investigated the convergence behavior for the molar volume and the length of the confined IL. As shown in Table 1, in the (20,20) carbon nanotube, the values for the molar

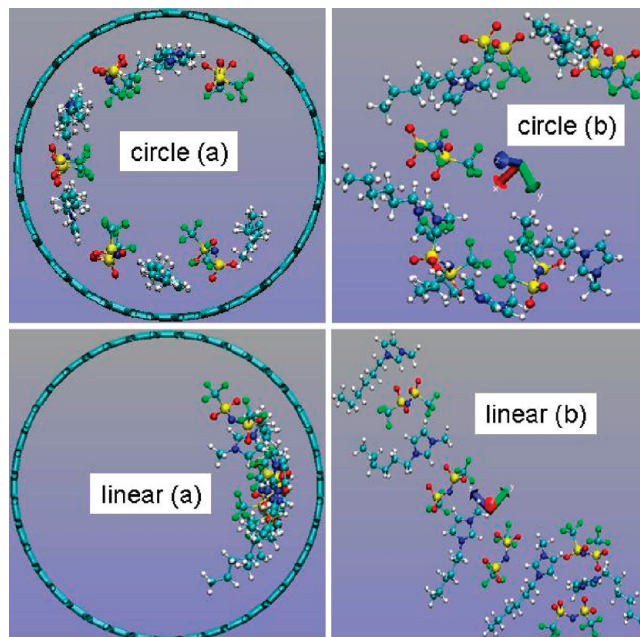


**Figure 1.** Simulated density versus temperature for the bulk ionic liquid at a pressure of 1 bar and the confined ionic liquid [hmim][Tf<sub>2</sub>N] at an axial pressure tensor  $P_{zz}$  of 1 bar. The circle symbols correspond to the bulk ionic liquid densities as obtained by Shi et al.<sup>25</sup> The squares denote the results of the ionic liquid confined inside the (20,20) carbon nanotube. The lines represent linear fitting of the simulated values.

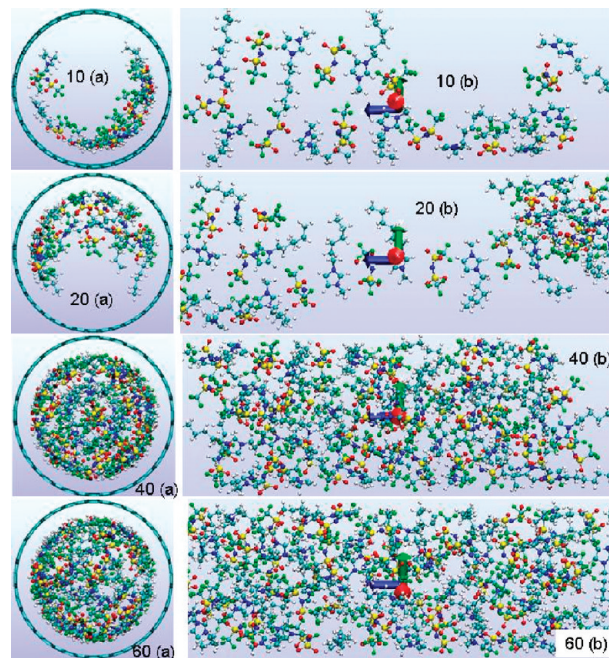
volume  $V_m$  and the height  $H_z$  are close among different simulations using different starting configurations and different initial  $H_z$  values. The largest difference in the molar volume and  $H_z$  values among different simulations is only about 3%. This good convergence strongly indicates that we have run long simulations to get reliable values. Snapshots (not shown here) from simulations also show that the ionic liquid molecules are closely packed to each other without any apparent big void spaces in the confined ionic liquid and that any void space which might be observed during simulations would disappear if long enough simulations were performed.

The simulated densities at different temperatures and axial pressure tensor  $P_{zz}$  of 1 bar for the confined IL and pressure of 1 bar for the bulk IL are shown in Figure 1. As shown in the figure, the density of the [hmim][Tf<sub>2</sub>N] confined in the (20,20) carbon nanotube decreases linearly with temperature from 313 to 573 K. This is similar to the behavior observed for the bulk ionic liquid. Interestingly, both the confined and the bulk ionic liquid exhibit similar density–temperature linear relationships. The slopes for the two lines are close to each other as shown in Figure 1. The density of the confined ionic liquid is smaller than the one for the bulk. As stated above, the radius of the carbon nanotube was used to compute the volume. If we define the volume as the accessible region inside the carbon nanotube to the ionic liquid, this will lead to a higher density for the confined ionic liquid. However, we have not investigated this further since it is not our focus in this work. Note that we have not systematically studied the system size effect on density for the confined IL but we expect that this effect will not be significant given the appreciable number of 60 ionic liquid pairs used in simulations.

**4.1.2. Structures of [hmim][Tf<sub>2</sub>N] Confined into CNT.** In the case when five cation–anion pairs of ionic liquid are adsorbed in the (20,20) CNT with a tube length of 95.921 Å, the ionic liquid molecules are always clustered to each other and no fragmentation of the ionic liquid structure was observed during a 20 ns trajectory. Two representative snapshots from this trajectory are shown in Figure 2. The ionic liquid molecules can form either a cylindrical-like configuration around the tube wall or a linear-like configuration along the tube wall. Both these two kinds of configurations are located close to the tube wall. The snapshots for 10, 20, 40, and 60 pairs of ionic liquids confined in the (20,20) CNT also exhibit similar behavior as for the 5 pairs. As shown in Figure 3, the ionic liquid molecules are clustered to each other at different loadings from 10 to 60



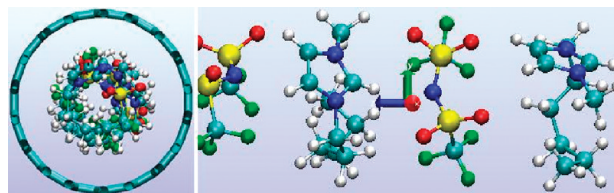
**Figure 2.** Two representative snapshots for 5 pairs of [hmim][Tf<sub>2</sub>N] adsorbed in (20,20) carbon nanotube with a tube length of 95.921 Å from NVT molecular dynamics simulations. The ionic liquid molecule can form either a cylindrical-like configuration around the carbon nanotube wall (circle (a) and circle (b)) or a linear-like configuration along the nanotube wall (linear (a) and linear (b)). The (a) and (b) configurations are viewed perpendicular to and along the carbon nanotube axis, respectively.



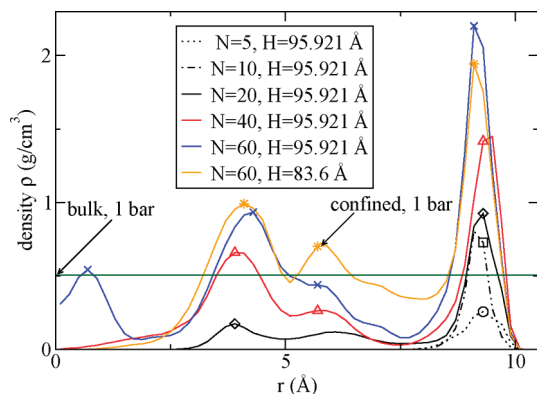
**Figure 3.** Representative snapshots for 10, 20, 40, and 60 pairs of [hmim][Tf<sub>2</sub>N] confined in the (20,20) carbon nanotube with a tube length of 95.921 Å. The (a) and (b) configurations are viewed perpendicular to and along the carbon nanotube axis, respectively.

pairs of IL. For the system of 10 pairs, the cations and anions are adsorbed close to the tube wall within a single layer. When the number of IL pairs is larger than 20, more layers farther away from the tube wall occur. In the case of (9,9) CNT, the 60 pairs of confined IL molecules also exhibit a linear-like configuration along the tube axis as shown in Figure 4. Note that we started the simulations by placing the cations and anions





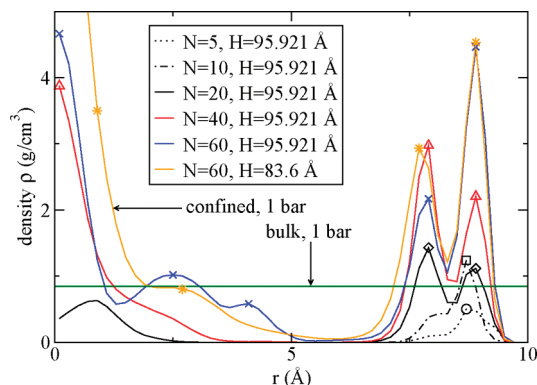
**Figure 4.** Representative snapshots of 60 pairs of [hmim][Tf<sub>2</sub>N] adsorbed inside the (9,9) carbon nanotube with a tube length of 597.661 Å. The left and right configurations are viewed perpendicular to and along the carbon nanotube axis, respectively.



**Figure 5.** Local density of the center of mass for the cation [hmim]<sup>+</sup> as a function of loading of the ionic liquid confined in the (20,20) carbon nanotube at 313 K. For comparison, the bulk cation density at 313 K and 1 bar is also shown as the green horizontal line. The dotted black line with circle symbols represents a system of 5 pairs of [hmim][Tf<sub>2</sub>N] confined in a carbon nanotube with a tube length  $H_z$  of 95.921 Å. The other lines are as follows: black dot-dash line with square symbols for 10 pairs and  $H_z$  of 95.921 Å; black solid line with diamond symbols for 20 pairs and  $H_z$  of 95.921 Å; red solid line with triangle symbols for 40 pairs and  $H_z$  of 95.921 Å; blue solid line with cross symbols for 60 pairs and  $H_z$  of 95.921 Å; brown solid line with star symbols for 60 pairs and  $H_z$  of 83.6 Å which corresponds to an axial pressure tensor  $P_{zz}$  of 1 bar. For clarity, different symbols are also shown on the corresponding lines.

inside the (9,9) CNT in an alternate succession. This is consistent with the fact that energetically it is more favorable for ions to enter inside the nanotube as cation–anion pairs as was shown to be the case for [bmim][PF<sub>6</sub>] adsorbed in the (9,9) CNT.<sup>15</sup> The radius of 6.102 Å for the (9,9) CNT is too small for the [hmim][Tf<sub>2</sub>N] molecules to form cylindrical-like configurations around the tube wall. Along the tube axis, the cation and the anion occur alternatively and the relative order remains the same as in the starting configuration. In contrast, in the case of (20,20) CNT, the radius of 13.56 Å is large enough to accommodate multiple layers of ionic liquid as shown in Figure 3.

The local density distribution of the center of mass for the cation [hmim]<sup>+</sup> is shown in Figure 5. Results in this case were obtained from the analysis of the trajectories generated from NVT MD simulations. Note that in these cases the local densities were computed exactly since the volume was defined without ambiguity as  $\pi(r_2^2 - r_1^2)H_z$ , where  $r_2$  and  $r_1$  are the outside and inside radii for a concentric cylindrical shell. In Figure 5, the tube center corresponds to  $r = 0$ . At the lowest loading of 5 and 10 pairs of ionic liquid, there is only one layer of cations adsorbed close to the tube wall. This is illustrated by the snapshots shown in Figures 2 and 3. The peak density corresponding to 5 pairs of IL denoted by circle symbols in Figure 5 is smaller than the bulk cation density at 1 bar. When the loading is increased to 10 pairs, there is still only one layer of cations adsorbed close to the wall. The peak density

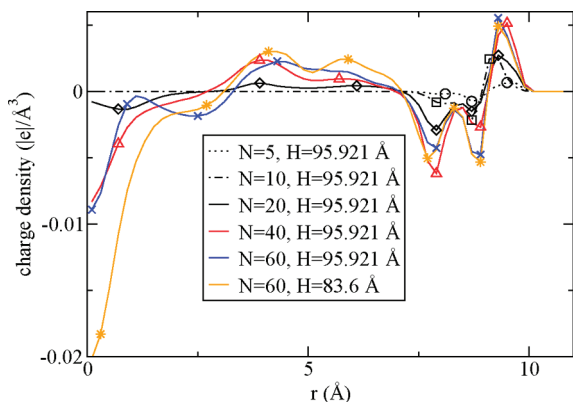


**Figure 6.** Local density of the center of mass for the anion [Tf<sub>2</sub>N]<sup>−</sup> at different loadings of the ionic liquid confined in the (20,20) carbon nanotube at 313 K. The symbols are the same as those in Figure 5.

corresponding to 10 pairs is higher than the one corresponding to 5 pairs, but the peak position occurs at almost the same location as for 5 pairs. Interestingly, at the low loading of 10 pairs, the peak density is higher than the bulk cation density. This suggests that the ionic liquid interacts very strongly with the (20,20) CNT, leading to an enhancement of the cation density close to the tube wall. When the loading is increased further above 20 pairs of IL, multiple layers occur at distances farther away from the tube wall. This is consistent with the snapshots shown in Figure 3. Typically, the peak densities for each layer increase and occur at almost the same position when the loading is increased. The cations in the layer close to the tube wall are more densely packed than those in other layers farther away from the tube wall. Overall, in the (20,20) carbon nanotube, the cations prefer to be adsorbed close to the tube wall rather than in the center as there are very few cations adsorbed in the tube center. This may be partly due to the hydrophobic characteristics for both the CNT and the long hexyl chain attached on the cation.

The local density profiles for the anion [Tf<sub>2</sub>N]<sup>−</sup> at different ionic liquid loadings in the (20,20) carbon nanotube are shown in Figure 6. They show similar behavior as the cations. At low loadings of 5 and 10 pairs, there is only one layer of anion adsorbed close to the tube wall. This is consistent with the snapshots shown in Figures 2 and 3. At higher loading larger than 20 pairs of IL, more layers positioned farther away from the tube wall occur, as can be seen from the snapshots illustrated in Figure 3. Additionally, as shown in Figure 6, for each layer, the peak height increases and occurs at almost the same position when the loading is increased. In contrast to the cations, when the loading is increased above 40 ion pairs in the (20,20) carbon nanotube, the anions prefer to adsorb close to the tube center rather than the tube wall, as shown in Figure 6 by the higher peaks in the center than those close to the wall.

We have also computed the charge distribution of the ionic liquid confined in the (20,20) carbon nanotube. This was obtained from the density distributions of the cations and anions shown in Figures 5 and 6 and the corresponding results are indicated in Figure 7. The charge distributions exhibit an oscillatory behavior. Close to the tube wall, there are more cations leading to an overall positive charge density at all loadings. In contrast, there are more anions in the tube center and the overall charge density is negative at loadings larger than 20 pairs of IL. Not surprisingly, the charge distribution is consistent with the cation and anion distribution shown in Figures 5 and 6, respectively. This suggests that by selecting different confinement materials, e.g., hydrophobic or hydrophilic

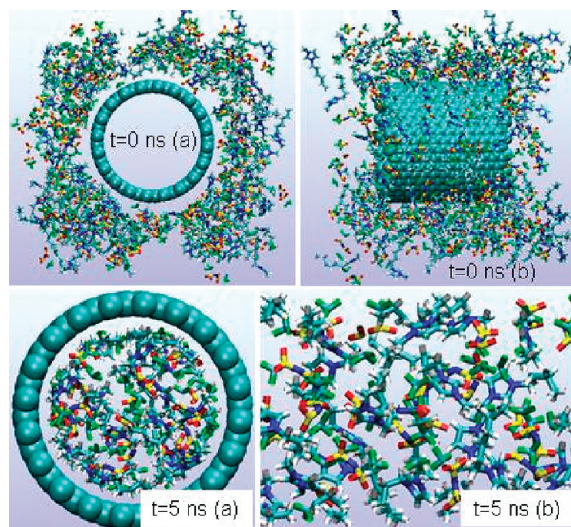


**Figure 7.** Charge density distribution for the ionic liquid [hmim][Tf<sub>2</sub>N] confined in the (20,20) carbon nanotube at 313 K and at different loadings. The symbols are the same as those in Figure 5.

nanoporous materials, the cation or anion will be preferentially adsorbed close to the wall or to the nanotube center. Such distributions are obviously different from those observed in bulk ILs. The confinement hence will affect the ionic liquid properties.

**4.1.3. Interaction Energies.** We have computed the interactions between the IL molecules themselves, and between the IL and the CNT. The corresponding results are shown in Table 2. As expected, the bonded energy  $U_m^b$  is independent of the loading. When the loading is increased, the solid–fluid interaction energy  $U_m^{sf}$  between the CNT and the IL becomes weaker and changes from  $-114.7$  kJ/mol for 5 pairs of ionic liquid to  $-80.0$  kJ/mol corresponding to 60 pairs. This is due to the fact that more layers positioned farther away from the tube wall will be formed when the loading is increased. The cations and anions in the layers farther away from the tube wall have weaker interactions with the CNT than those close to the tube wall. In the case of the systems with 5 and 10 pairs, as discussed above, there is only one layer of cations and anions as shown in Figures 5 and 6. This leads to very strong solid–fluid interaction energies  $U_m^{sf}$  as shown in Table 2. At all loadings, the electrostatic interaction between the ionic liquid molecules is small and the van der Waals interaction energy between the ionic liquids and the solid–ionic liquid interaction predominates over the electrostatic energy. Especially at low loading, the solid–fluid interaction is the most significant.

**4.1.4. Transfer of the IL from the Bulk Phase Inside CNT.** Note that in MC and MD calculations, simulations were started from initial configurations where the ionic liquid molecules are already adsorbed inside the carbon nanotube. To investigate whether the [hmim][Tf<sub>2</sub>N] molecules could penetrate inside the CNT from the bulk phase, we performed additional NPT MD



**Figure 8.** Two snapshots from NPT MD simulations for ionic liquid [hmim][Tf<sub>2</sub>N] molecules transferred from the bulk phase inside the (20,20) carbon nanotube at 0 and 5 ns, respectively. The (a) and (b) configurations are viewed perpendicular to and along the carbon nanotube axis, respectively. For clarity reasons, at 5 ns, only 20 pairs of [hmim][Tf<sub>2</sub>N] inside the carbon nanotube are shown and the carbon nanotube itself is not shown in panel (b).

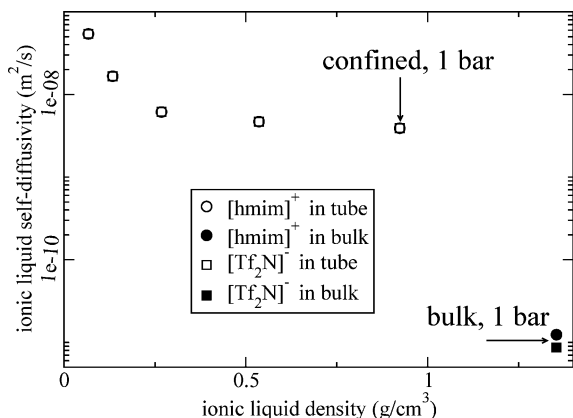
simulation at 573 K and 1 bar with 130 pairs of [hmim][Tf<sub>2</sub>N]. The simulations were started from an empty (20,20) carbon nanotube without IL molecules inside. After 2 ns, about 12 pairs of IL molecules are transferred inside the CNT. This number is increased to about 20 pairs of IL molecules at 5 ns. Snapshots from simulations at the starting and after 5 ns are shown in Figure 8. This suggests that the interaction between the carbon nanotube and the [hmim][Tf<sub>2</sub>N] is strong enough to overcome the free energy penalty required for transferring IL molecules from the bulk phase inside the CNT. This strong interaction between the CNT and the IL is also evidenced by the large  $U_m^{sf}$  values indicated in Table 2. These results (see Figure 8) suggest that experimentally one can easily obtain the composite material by sinking the opened carbon nanotubes into the [hmim][Tf<sub>2</sub>N] ionic liquid.

**4.2. Self-Diffusivity Coefficients.** **4.2.1. Confined IL in CNT.** Self-diffusion coefficients  $D_{self,z}$  for the IL in the Z direction of the CNT were calculated from NVE MD trajectories. In order to confirm the calculation procedure using the NAMD program,<sup>29</sup> we have computed the self-diffusion coefficients for bulk CH<sub>4</sub> and CH<sub>4</sub> adsorbed into the (15,0) CNT using both our in-house-developed MD code and the NAMD software. The simulation results from both software packages agree with each other as well as with the ones obtained by other researchers.<sup>31,36</sup> As we observed in our previous work,<sup>25</sup> the

**TABLE 2: Specific Total Energy  $U_m$ , Bonded Energy  $U_m^b$ , Nonbonded Energy  $U_m^{nb}$ , and the van der Waals  $U_m^{vdw}$  and the Electrostatic  $U_m^{Ewald}$  Energies between Ionic Liquid Molecules, and the Interaction Energy  $U_m^{sf}$  between the Carbon Nanotube and the Ionic Liquid Molecules<sup>a</sup>**

| no. of pairs | $H_z$ (Å) | $U_m$ (kJ/mol) | $U_m^b$ (kJ/mol) | $U_m^{nb}$ (kJ/mol) | $U_m^{vdw}$ (kJ/mol) | $U_m^{sf}$ (kJ/mol) | $U_m^{Ewald}$ (kJ/mol) |
|--------------|-----------|----------------|------------------|---------------------|----------------------|---------------------|------------------------|
| 5            | 95.921    | 13.9 (4)       | 160.7 (1)        | -146.9 (3)          | -31.2 (3)            | -114.7 (5)          | -1.0 (5)               |
| 10           | 95.921    | 2.0 (6)        | 160.1 (1)        | -158.1 (7)          | -31.5 (4)            | -116.8 (4)          | -9.8 (7)               |
| 20           | 95.921    | -2.7 (6)       | 161.4 (1)        | -164.0 (6)          | -49.1 (4)            | -94.3 (6)           | -20.6 (6)              |
| 40           | 95.921    | -14.5 (4)      | 161.5 (1)        | -176.0 (4)          | -67.9 (4)            | -82.0 (2)           | -26.1 (4)              |
| 60           | 95.921    | -17.1 (1)      | 161.1 (1)        | -178.2 (1)          | -71.3 (1)            | -80.0 (1)           | -26.9 (1)              |
| 60           | 83.624    | -15.4 (2)      | 161.6 (1)        | -177.0 (2)          | -75.6 (1)            | -72.5 (1)           | -28.9 (3)              |

<sup>a</sup> The  $U_m^{nb}$  consists of  $U_m^{vdw}$ ,  $U_m^{Ewald}$ , and  $U_m^{sf}$ . The energies were computed from NVT molecular dynamics simulations at 313 K. The specific energies were calculated from the corresponding extensive ones divided by the number of pairs of the cations and anions. The uncertainty in the last digit is given in parentheses.



**Figure 9.** Simulated self-diffusivity values for [hmim][Tf<sub>2</sub>N] confined in the (20,20) carbon nanotube at 313 K and different loadings. The loadings correspond to 5, 10, 20, and 40 pairs of ionic liquid molecules adsorbed inside the nanotube with a tube length of 95.921 Å, and to 60 pairs inside the tube with a tube length of 83.623 Å, which roughly correspond to an axial pressure tensor  $P_{zz}$  of 1 bar. For comparison, the self-diffusivity values for the bulk ionic liquid at 313 K and a pressure of 1 bar are also shown. The error bars are smaller than the symbols and are not shown.

Langevin NVT MD simulations underestimate the self-diffusivity. The Langevin NVT MD gives self-diffusivity values for the bulk CH<sub>4</sub> about 3–40 times smaller than those obtained from NVE MD simulations at room temperature and at different densities. The damping coefficient in the Langevin NVT simulations was set to be 5 ps<sup>-1</sup>. We have not systematically studied the damping coefficient effect on the self-diffusivity since it is not our focus in this work. Hence, in order to get reliable self-diffusivity data, one needs to perform NVE MD simulations. The simulated self-diffusivity values for the [hmim][Tf<sub>2</sub>N] inside the (20,20) CNT at 313 K and at different loadings are shown in Figure 9. The self-diffusivity values decrease with loading due to the increased interaction between the ionic liquid molecules as shown in Table 2. However, we note that the decrease in the self-diffusivity for the confined IL with loading is gradual. For example, the self-diffusivity of the ionic liquid inside the tube decreases by about 1 order of magnitude when the loading is increased from 5 pairs to 60 pairs. In contrast, the self-diffusivity for nonionic liquid molecules such as simple CH<sub>4</sub> molecules decreases more abruptly when the loading is increased.<sup>31</sup> This is due to the fact that CH<sub>4</sub> molecules are not clustered to each other at low loading. Instead they are evenly distributed along the tube axis. The interaction between the CH<sub>4</sub> molecules is negligibly small at low loading which leads to large self-diffusivity. In contrast, for the ionic liquid even at a low loading of 5 pairs, the IL molecules are clustered to each other (see Figure 2) and the interaction between IL molecules is significant (see Table 2). This leads to somewhat low self-diffusivity for the ionic liquid at low loading compared to the case of simple CH<sub>4</sub> molecules.

Another very interesting observation is that the ionic liquid molecules inside the (20,20) carbon nanotube diffuse about 300 times faster than the bulk ionic liquid at the same temperature of 313 K and pressure of 1 bar. We have also computed the self-diffusivity for 60 pairs of [hmim][Tf<sub>2</sub>N] confined in the (9,9) carbon nanotube with a tube length of 597.661 Å at 313 K. This length roughly corresponds to the confined IL density inside the (9,9) CNT at 1 bar and 313 K as shown in Table 1. The simulated self-diffusivity values for the cation and anion are about 350 times faster in the CNT than in the bulk ionic liquid. At 1 bar and higher temperatures of 423 and 573 K, the

**TABLE 3: Self-Diffusivity Values in the Z Direction along the Nanotube Axis for CO<sub>2</sub> and H<sub>2</sub> at 313 K<sup>a</sup>**

| system                                      | $D_{\text{self},z}(\text{CO}_2)$ (m <sup>2</sup> /s) | $D_{\text{self},z}(\text{H}_2)$ (m <sup>2</sup> /s) |
|---------------------------------------------|------------------------------------------------------|-----------------------------------------------------|
| gas/CNT                                     | $2.5(5) \times 10^{-6}$                              | $3.3(3) \times 10^{-5}$                             |
| gas/composite material                      | $4.4(3) \times 10^{-9}$                              | $6.7(1) \times 10^{-9}$                             |
| gas/[hmim][Tf <sub>2</sub> N] <sup>25</sup> | $2.59(4) \times 10^{-10}$                            | $3.15(6) \times 10^{-9}$                            |

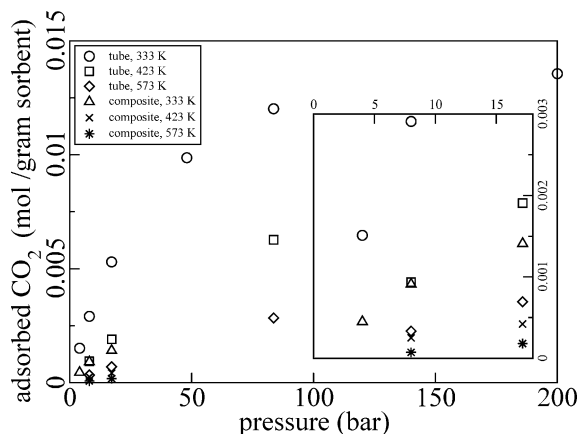
<sup>a</sup> The gas/carbon nanotube system consists of 30 gas molecules (CO<sub>2</sub> or H<sub>2</sub>) adsorbed in a (20,20) carbon nanotube with a tube length of 836.234 Å. The gas/[hmim][Tf<sub>2</sub>N] system consists of five gas molecules adsorbed in 160 pairs of [hmim][Tf<sub>2</sub>N], and the results from the ref 25 are included here. The gas/composite material contains three gas molecules and 60 pairs of [hmim][Tf<sub>2</sub>N] confined in a (20,20) carbon nanotube with a tube length of 83.623 Å. The uncertainty in the last digit is given in parentheses.

self-diffusivity coefficients for the confined [hmim][Tf<sub>2</sub>N] in the (20,20) CNT were calculated to be  $(8.4 \pm 0.5) \times 10^{-9}$  and  $(1.7 \pm 0.1) \times 10^{-8}$  m<sup>2</sup>/s, respectively. These diffusivity values for the confined [hmim][Tf<sub>2</sub>N] are about 41 and 14 times larger than the corresponding bulk ionic liquid self-diffusivities.

The fast self-diffusivity for ILs in carbon nanotube is partly due to the weaker interactions among the ions under confinement. For example, for the bulk IL [hmim][Tf<sub>2</sub>N] at 333 K and 48 bar, the van der Waals and electrostatic interaction energies per cation–anion pair were computed to be  $-102.29 \pm 0.08$  and  $-34.55 \pm 0.08$  kJ/mol, respectively. In contrast, in the case of IL confined in the (20,20) CNT, the corresponding van der Waals and the electrostatic interaction energies among the ions per cation–anion pair were found to be  $-73.1 \pm 0.3$  and  $-26.4 \pm 0.2$  kJ/mol, respectively, at the same temperature and pressure condition. The van der Waals and electrostatic interaction energies are about 25–30% weaker in the confinement than in the bulk IL. In turn, this weaker interaction under confinement partly contributes to the fast self-diffusion. In the case of (20,20) nanotube, we have also computed a radius-dependent self-diffusivity by analyzing the ILs close to the tube wall and in the center. As shown in Figures 5 and 6, the layer close to the tube wall was chosen in the region  $7.3 \text{ Å} \leq r \leq 10.5 \text{ Å}$  for the cation and  $5.7 \text{ Å} \leq r \leq 10.5 \text{ Å}$  for the anion, while the layer in the tube center was chosen in the region  $0.0 \text{ Å} \leq r \leq 7.3 \text{ Å}$  for the cation and  $0.0 \text{ Å} \leq r \leq 5.7 \text{ Å}$  for the anion. For both the cations and the anions, the self-diffusivity values in the layer close to the tube wall are about 10% smaller than those in the layer close to the tube center. This is expected since the ILs close to the tube wall are typically more densely packed than those in the tube center. Despite the structure differences, we notice that the self-diffusivities of the cation and the anion confined in the carbon nanotube are very close to each other. These findings indicate that there exist strong coordinated moves of the cations and the anions due to strong attractive electrostatic interactions between them.

**4.2.2. CO<sub>2</sub> and H<sub>2</sub> Adsorbed in CNT and in Composite CNT–IL Material.** The self-diffusivity coefficients for CO<sub>2</sub> and H<sub>2</sub> gases have been calculated at 313 K for two systems (see Table 3). In the first case, 30 gas (CO<sub>2</sub> or H<sub>2</sub>) molecules are adsorbed inside a (20,20) carbon nanotube with a tube length of 836.234 Å, corresponding to low loadings. In the second case, three gas (CO<sub>2</sub> or H<sub>2</sub>) molecules plus 60 pairs of [hmim][Tf<sub>2</sub>N] are adsorbed inside the (20,20) carbon nanotube with a tube length of 83.623 Å. This height corresponds to the confined [hmim][Tf<sub>2</sub>N] density in the (20,20) CNT at 313 K and 1 bar as shown in Table 1. The first and second cases correspond to the same gas (CO<sub>2</sub> or H<sub>2</sub>) loading of 1 gas molecule/27.874 Å height of (20,20) CNT. For both CO<sub>2</sub> and H<sub>2</sub>, the gases exhibit the highest diffusivity inside the tube,



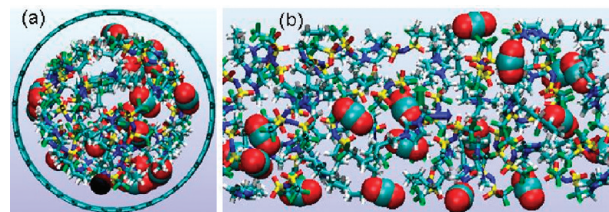


**Figure 10.** Isotherms of CO<sub>2</sub> adsorbed inside the (20,20) carbon nanotube, and in the composite material consisting of 60 pairs of [hmim][Tf<sub>2</sub>N] and a (20,20) carbon nanotube at 333, 423, and 573 K, and at different pressures/axial pressure tensors. The error bars are smaller than the symbols and are not shown. For clarity, the inset panel indicates the simulated results at low pressures/axial pressure tensors.

followed by the composite material, and the gases in the ionic liquid show the smallest self-diffusivities. Although H<sub>2</sub> diffuses about 10 times faster than CO<sub>2</sub> in both the tube and in the ionic liquid, the difference between diffusivities of the two gases is small in the composite material. The H<sub>2</sub> diffuses only about 1.5 times faster than CO<sub>2</sub> in the composite material.

**4.3. Sorption Isotherms of CO<sub>2</sub> and H<sub>2</sub> in the Composite Material and in CNT.** **4.3.1. CO<sub>2</sub> Sorption.** The isotherms for CO<sub>2</sub> adsorbed into the (20,20) CNT and in the composite material are shown in Figure 10. The amount of gas adsorbed was calculated as the number of moles of CO<sub>2</sub> divided by the mass of the tube or the mass of the composite material (total mass of the IL and of the CNT). At the same temperature and pressure, the CO<sub>2</sub> adsorbed in the (20,20) CNT is about 3–4 times larger than in the composite material. This is partly due to the large void space inside the (20,20) CNT available for CO<sub>2</sub> sorption than in the composite material. When compared with CO<sub>2</sub> absorbed in the ionic liquid [hmim][Tf<sub>2</sub>N], the amount of CO<sub>2</sub> in the composite material is about 2–3 times larger than that in the bulk IL. This is partly due to the stronger interactions between the CO<sub>2</sub> and the CNT than those between the CO<sub>2</sub> and the IL, as will be discussed later. As shown in Figure 11, at low pressure of 4 bar and 333 K, many CO<sub>2</sub> molecules are adsorbed close to the tube wall. This implies that the CNT interacts strongly with CO<sub>2</sub>.

As shown in Figure 10, the amount of adsorbed CO<sub>2</sub> increases linearly with pressure at low pressures. On the basis of a linear fit of the fugacity vs the amount of adsorbed CO<sub>2</sub> at low pressures, we have estimated the Henry's law constant (*H*) and the corresponding results are shown in Table 4. It can be observed that the *H* values increase with temperature for CO<sub>2</sub> in both the CNT and in the composite material.



**Figure 11.** Snapshot from CFC isostress–osmotic simulations for CO<sub>2</sub> molecules absorbed in the composite material consisting of 60 pairs of [hmim][Tf<sub>2</sub>N] and a (20,20) carbon nanotube at 333 K and 4 bar. The CO<sub>2</sub> molecules are indicated using the vdw graphical representation and the ionic liquid molecules are shown using the Bonds graphical representation. The (a) and (b) configurations are viewed perpendicular to and along the carbon nanotube axis, respectively. In panel (b), the (20,20) carbon nanotube itself is not shown for clarity.

By plotting log(*H*) vs 1/*T* and performing a linear regression, the enthalpy values for CO<sub>2</sub> adsorbed in the (20,20) CNT and in the composite material were estimated to be  $-14.7 \pm 0.5$  and  $-15.9 \pm 2.3$  kJ/mol, respectively. These enthalpy values are more negative than the value of  $-10.3 \pm 1.4$  kJ/mol<sup>22</sup> for CO<sub>2</sub> absorption in [hmim][Tf<sub>2</sub>N], consistent with larger amounts of adsorbed CO<sub>2</sub> in the (20,20) CNT and in the composite material than in the bulk IL. The calculated amounts of adsorbed CO<sub>2</sub> in the CNT and in the composite material are summarized in Table 5 and Table 6, respectively.

The CO<sub>2</sub> sorption behavior in the ionic liquid is very different than that in the composite material. When CO<sub>2</sub> molecules are absorbed in [hmim][Tf<sub>2</sub>N] at temperatures higher than the bulk CO<sub>2</sub> critical temperature of about 304 K, the amount of adsorbed CO<sub>2</sub> was found to increase with pressure without liquid phase separation. It has been found that the ionic liquid always behaves as a solvent and no fragmentation was observed. For example, even at a high pressure of 200 bar and 333 K, the CO<sub>2</sub> solubility in the ionic liquid is about 0.7.<sup>22</sup> At this high CO<sub>2</sub> concentration, many CO<sub>2</sub> molecules are close to each other. However, the ionic liquid structure is not broken as a whole and the distance between the center of masses of the cation and the anion just expands a little bit compared to the pure ionic liquid without CO<sub>2</sub> absorption.<sup>22</sup> Consequently, CO<sub>2</sub> always behaves as a solute and the ionic liquid as a solvent.

In contrast, in the case of CO<sub>2</sub> absorbed in the [hmim][Tf<sub>2</sub>N] confined in the (20,20) CNT, the ionic liquid does not always behave as a continuous solvent. At a low pressure of 4 bar and 333 K (see Figure 11), the CO<sub>2</sub> molecules behave as solutes dissolved in the ionic liquid. However, when the pressure is increased above a certain value, for example at 48 bar and 333 K, the [hmim][Tf<sub>2</sub>N] molecules start to dissolve in the CO<sub>2</sub> phase. In this case, the CO<sub>2</sub> molecule behaves as a solvent and the ionic liquid behaves as a solute. As shown in Figure 12, the ionic liquid molecules can dissolve in CO<sub>2</sub> as a single ion (1), as an ion pair (2), as three ions (3), and as a cluster consisting of more than five ions. The IL no longer represents

**TABLE 4: Variation of Henry's Law Constant (*H*) as a Function of Temperature for CO<sub>2</sub> and H<sub>2</sub> Adsorbed into the (20,20) Carbon Nanotube and in a Composite Material Consisting of [hmim][Tf<sub>2</sub>N] and the (20,20) Carbon Nanotube<sup>a</sup>**

| <i>T</i> (K) | <i>H</i> (bar mol <sup>-1</sup> g) |                            |                     |                           |
|--------------|------------------------------------|----------------------------|---------------------|---------------------------|
|              | CO <sub>2</sub> /CNT               | CO <sub>2</sub> /composite | H <sub>2</sub> /CNT | H <sub>2</sub> /composite |
| 313          | —                                  | —                          | 29072 ± 1513        | 516085 ± 4043             |
| 333          | 2664 ± 34                          | 8521 ± 131                 | —                   | —                         |
| 373          | —                                  | —                          | 37559 ± 1563        | 683414 ± 37424            |
| 423          | 8746 ± 176                         | 39076 ± 5505               | —                   | —                         |
| 573          | 24562 ± 362                        | 93820 ± 7343               | 65749 ± 1538        | 558545 ± 12030            |

<sup>a</sup> The error bars are also indicated.

**TABLE 5:** Calculated Average Number  $\langle N \rangle$  of CO<sub>2</sub> and H<sub>2</sub> Molecules Adsorbed in the (20,20) Carbon Nanotube at Different Temperatures  $T$ , Pressures  $P$ , and Carbon Nanotube Lengths  $H_z$  from CFC Grand Canonical Simulations<sup>a</sup>

| system          | $T$ (K) | $f$ (bar) | $P$ (bar) | $H_z$ (Å) | $\langle N \rangle$ |
|-----------------|---------|-----------|-----------|-----------|---------------------|
| CO <sub>2</sub> | 333.2   | 3.9384    | 4         | 30        | 17.70 (0.07)        |
| CO <sub>2</sub> | 333.2   | 7.7552    | 8         | 30        | 34.1 (0.1)          |
| CO <sub>2</sub> | 333.2   | 16.04     | 17.15     | 30        | 62.1 (0.2)          |
| CO <sub>2</sub> | 333.2   | 39.6144   | 48.0      | 30        | 115.6 (0.6)         |
| CO <sub>2</sub> | 333.2   | 59.22     | 83.6      | 30        | 140.9 (0.3)         |
| CO <sub>2</sub> | 333.2   | 90.04     | 200       | 30        | 158.9 (0.6)         |
| CO <sub>2</sub> | 423.2   | 7.8936    | 8         | 60        | 21.95 (0.04)        |
| CO <sub>2</sub> | 423.2   | 16.66637  | 17.15     | 60        | 44.7 (0.1)          |
| CO <sub>2</sub> | 423.2   | 73.09148  | 83.6      | 60        | 146.9 (0.4)         |
| CO <sub>2</sub> | 573.2   | 7.976     | 8         | 120       | 15.65 (0.03)        |
| CO <sub>2</sub> | 573.2   | 17.04024  | 17.15     | 60        | 16.28 (0.03)        |
| CO <sub>2</sub> | 573.2   | 81.29264  | 83.6      | 60        | 66.6 (0.2)          |
| H <sub>2</sub>  | 313.2   | 51.3      | 50        | 25        | 18.79 (0.02)        |
| H <sub>2</sub>  | 313.2   | 105.3     | 100       | 25        | 35.31 (0.05)        |
| H <sub>2</sub>  | 313.2   | 221.8     | 200       | 25        | 63.19 (0.07)        |
| H <sub>2</sub>  | 313.2   | 350.7     | 300       | 25        | 86.3 (0.2)          |
| H <sub>2</sub>  | 313.2   | 492.8     | 400       | 25        | 106.3 (0.1)         |
| H <sub>2</sub>  | 373.2   | 51.2      | 50        | 25        | 14.28 (0.03)        |
| H <sub>2</sub>  | 373.2   | 104.8     | 100       | 25        | 27.22 (0.07)        |
| H <sub>2</sub>  | 373.2   | 219.4     | 200       | 25        | 50.32 (0.06)        |
| H <sub>2</sub>  | 373.2   | 344.7     | 300       | 25        | 69.99 (0.06)        |
| H <sub>2</sub>  | 573.2   | 50.6      | 50        | 50        | 15.65 (0.02)        |
| H <sub>2</sub>  | 573.2   | 102.8     | 100       | 25        | 15.27 (0.03)        |
| H <sub>2</sub>  | 573.2   | 213.2     | 200       | 25        | 29.38 (0.07)        |
| H <sub>2</sub>  | 573.2   | 330.6     | 300       | 25        | 42.38 (0.06)        |

<sup>a</sup> Also shown in the table are the fugacity  $f$  values obtained from Peng–Robinson equation of state for CO<sub>2</sub><sup>22</sup> and from molecular dynamics simulations for H<sub>2</sub>.<sup>25</sup> The uncertainty in the last digit is given in parentheses.

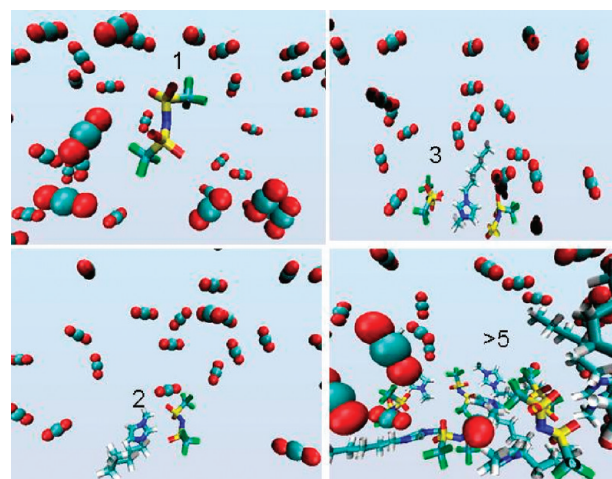
a solvent in which CO<sub>2</sub> is dissolved. Instead, it dissolves in the CO<sub>2</sub> and CO<sub>2</sub> behaves as the solvent. This effect was also observed at two other temperatures of 423 and 573 K under confinement conditions when IL molecules start to dissolve in CO<sub>2</sub> at about 83.6 bar.

**4.3.2. Energetic Requirement for Transferring an IL Pair from IL–CNT to CO<sub>2</sub>–CNT under Confinement.** To further understand ionic liquid solvation in CO<sub>2</sub> under confinement, we have computed the required energy of transferring one pair of IL from the IL phase to the CO<sub>2</sub> phase when both these two phases are confined in CNT. A generic scheme which illustrates this process is shown in Figure 13. In the case of confinement we have performed two sets of MD simulations. The first one is an NVT MD simulation of 60 pair of IL molecules confined in the (20,20) CNT with a tube length of about 83.623 Å at 333 K. This tube length roughly corresponds to the IL density confined in the (20,20) CNT at 333 K and 48 bar as obtained from NP<sub>zz</sub>T Monte Carlo simulations. The second NVT MD simulation was performed at the same  $T = 333$  K for a system consisting of one pair of IL plus 463 CO<sub>2</sub> molecules confined in the (20,20) CNT with a tube length of about 118.057 Å. The number of CO<sub>2</sub> molecules and the tube length in the second simulation correspond to the adsorbed CO<sub>2</sub> density in the (20,20) CNT at 333 K and 48 bar as shown in Table 5. Note that the bonded energies  $U_{\text{b}}^{\text{b}}$  consisting of bond, angle, dihedral, and improper potential terms before and after transfer of one pair of IL are very close to each other. Hence, we only evaluated the nonbonded energy  $U_{\text{nb}}^{\text{b}}$  when computing the transfer energy. The  $U_{\text{nb}}^{\text{b}}$  values for one pair of IL in the first and second runs are  $-173.2 \pm 0.2$  kJ/mol, and  $-181.6 \pm 2.5$  kJ/mol, respectively, leading to a transferring energy of  $-8.4 \pm 2.5$  kJ/mol

**TABLE 6:** Computed Average Number  $\langle N \rangle$  of CO<sub>2</sub> and H<sub>2</sub> Molecules Sorbed in the Composite Material Consisting of a (20,20) Carbon Nanotube and 60 Pairs of [hmim][Tf<sub>2</sub>N] Molecules at Different Temperatures  $T$ , Axial Pressure Tensors  $P_{zz}$ , and Fugacities  $f$  from CFC Isostress–Osmotic Simulations<sup>a</sup>

| system          | $T$ (K) | $f$ (bar) | $P$ (bar) | $H_z$ (Å)    | $\langle N \rangle$ |
|-----------------|---------|-----------|-----------|--------------|---------------------|
| CO <sub>2</sub> | 333.2   | 3.9384    | 4         | 86.9 (0.1)   | 27.1 (1.0)          |
| CO <sub>2</sub> | 333.2   | 7.7552    | 8         | 93.0 (0.3)   | 57.3 (2.0)          |
| CO <sub>2</sub> | 333.2   | 16.04     | 17.15     | 96.4 (0.1)   | 91.1 (1.2)          |
| CO <sub>2</sub> | 423.2   | 7.8936    | 8         | 93.5 (0.3)   | 15.6 (0.8)          |
| CO <sub>2</sub> | 423.2   | 16.66637  | 17.15     | 93.8 (0.4)   | 26.5 (1.2)          |
| CO <sub>2</sub> | 573.2   | 7.976     | 8         | 104.7 (0.4)  | 4.9 (0.3)           |
| CO <sub>2</sub> | 573.2   | 17.04024  | 17.15     | 111.0 (0.4)  | 12.7 (0.3)          |
| H <sub>2</sub>  | 313.2   | 51.3      | 50        | 82.8 (0.2)   | 5.8 (0.1)           |
| H <sub>2</sub>  | 313.2   | 105.3     | 100       | 83.12 (0.06) | 12.1 (0.2)          |
| H <sub>2</sub>  | 313.2   | 221.8     | 200       | 83.1 (0.2)   | 22.1 (0.4)          |
| H <sub>2</sub>  | 313.2   | 350.7     | 300       | 82.0 (0.2)   | 26.3 (0.8)          |
| H <sub>2</sub>  | 313.2   | 492.8     | 400       | 82.2 (0.1)   | 35.2 (0.7)          |
| H <sub>2</sub>  | 373.2   | 51.2      | 50        | 85.9 (0.1)   | 5.0 (0.1)           |
| H <sub>2</sub>  | 373.2   | 104.8     | 100       | 85.9 (0.2)   | 9.2 (0.2)           |
| H <sub>2</sub>  | 373.2   | 219.4     | 200       | 85.1 (0.2)   | 16.6 (0.5)          |
| H <sub>2</sub>  | 373.2   | 344.7     | 300       | 86.3 (0.3)   | 28.0 (0.6)          |
| H <sub>2</sub>  | 573.2   | 50.6      | 50        | 103.7 (0.3)  | 6.36 (0.09)         |
| H <sub>2</sub>  | 573.2   | 102.8     | 100       | 103.4 (0.6)  | 12.4 (0.3)          |
| H <sub>2</sub>  | 573.2   | 213.2     | 200       | 102.7 (0.1)  | 23.5 (0.2)          |
| H <sub>2</sub>  | 573.2   | 330.6     | 300       | 104.9 (0.9)  | 38.3 (1.6)          |

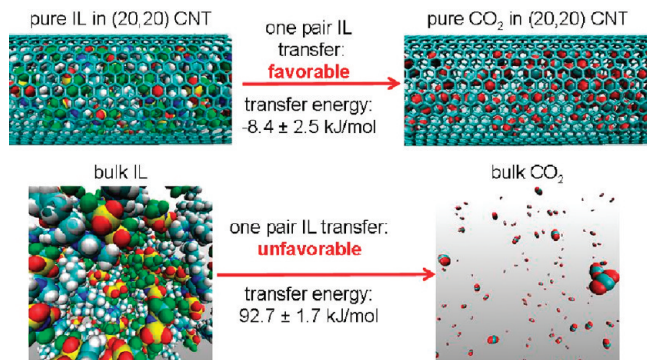
<sup>a</sup> Also shown are the simulated results of the average length  $H_z$  for the ionic liquid column in the tube axis direction upon gases absorption. Fugacities for CO<sub>2</sub> and H<sub>2</sub> were computed from the Peng–Robinson equation of state<sup>22</sup> and molecular dynamics simulation,<sup>25</sup> respectively. The uncertainty in the last digit is given in parentheses.



**Figure 12.** Representative snapshots for [hmim][Tf<sub>2</sub>N] ionic liquid dissolved in the adsorbed CO<sub>2</sub> in the (20,20) carbon nanotube at 333 K and 48 bar from CFC isostress–osmotic simulations. For clarity reasons, the carbon nanotube is not shown. The number 1 denotes one cation or one anion, the number 2 denotes one pair of [hmim][Tf<sub>2</sub>N] molecule, and so on.

(see Figure 13). These results imply that it is energetically favorable to transfer one pair of IL from the IL phase to the CO<sub>2</sub> phase under confinement at 333 K and 48 bar. Under these conditions, the IL molecules dissolve in the CO<sub>2</sub> phase as shown in Figure 12. Note that a more accurate way of evaluating this transfer process is based on free energy calculations. However, computing the transfer free energy is a nontrivial task and it is outside of the scope of this work. Hence, we use the transferring energy as an approximation of the transferring free energy.

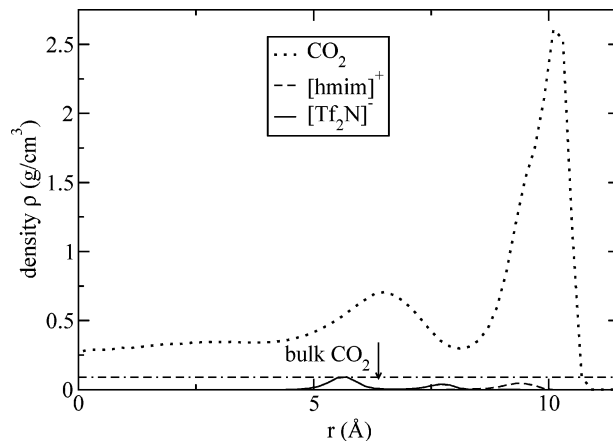
In the case of transferring one pair of IL from the bulk IL phase to the bulk CO<sub>2</sub> phase without confinement, as shown in



**Figure 13.** Schematic transfer of one pair of [hmim][Tf<sub>2</sub>N] from the confined ionic liquid to the confined CO<sub>2</sub>, and from the bulk (unconfined) ionic liquid to the bulk (unconfined) CO<sub>2</sub>. Also shown are the corresponding transfer energy values.

the bottom part of Figure 13, we have also performed two sets of simulations. The first set represents an NPT MD simulation performed at 333 K and 48 bar for 160 pairs of [hmim][Tf<sub>2</sub>N], while the second set is an NVT MD simulation for a system containing one pair of IL plus 463 CO<sub>2</sub> molecules in a cubic box with a side of about 71.979 Å. The box size in the second run corresponds to the simulated bulk CO<sub>2</sub> molar volume of  $485.0 \pm 0.6$  cm<sup>3</sup>/mol at 333 K and 48 bar, value which is also close to the experimental one of 471.5 cm<sup>3</sup>/mol.<sup>37</sup> Similar to the confinement case, the  $U_m^b$  values are close to each other in the two runs and were not included in the transferring energy. The  $U_m^{nb}$  values in the first and second runs were calculated to be  $-136.9 \pm 0.2$  and  $-44.2 \pm 1.7$  kJ/mol, respectively, which leads to a transferring energy of  $92.7 \pm 1.7$  kJ/mol. This result indicates that energetically it is very unfavorable to transfer one pair of IL from the IL phase to the bulk CO<sub>2</sub> phase without confinement.

For the above two cases of one pair of IL plus 463 CO<sub>2</sub> molecules with and without confinement, we have decomposed the interaction energy between one pair of IL and the rest of the system into the following parts, i.e., cation–CO<sub>2</sub>, anion–CO<sub>2</sub>, cation–CNT, and anion–CNT. All these interaction energy values were computed<sup>38</sup> based on a direct summation of the van der Waals (vdw) and electrostatic energies between the respective species. In the summation, we have used a primary cell plus 124 imaging cells. The energy results from these 125 cells are consistent with those obtained in simulations for a larger system, corresponding to 343 cells, suggesting that we have used large enough cells to compute the interaction energy between different molecule species. We have also computed the self-interaction energy of one pair of IL denoted as the cation–anion energy and the corresponding results are indicated in Table 7. Note that, in both these two cases with and without confinement, the anion–CO<sub>2</sub> interaction is stronger than that between cation and CO<sub>2</sub>. Interestingly, the vdw energy part is stronger and more negative than the electrostatic one between the IL and CO<sub>2</sub>. The self-interaction energies were found to be



**Figure 14.** Local density for the center of mass of the cation [hmim]<sup>+</sup>, the anion [Tf<sub>2</sub>N]<sup>−</sup>, and CO<sub>2</sub> molecules for a system consisting of one pair of ionic liquid molecule and 463 CO<sub>2</sub> molecules adsorbed in a (20,20) carbon nanotube with a tube length of 118.057 Å at 333 K. The density of this system was selected to be equal to the one for CO<sub>2</sub> adsorbed in the (20,20) carbon nanotube at 333 K and a pressure of about 48 bar as shown in Table 5. For comparison, the simulated density for the bulk CO<sub>2</sub> at 333 K and 48 bar is also shown as the horizontal line. The dotted line corresponds to the density of adsorbed CO<sub>2</sub>. The dashed and solid lines indicate the density of the adsorbed cation and the anion, respectively.

close to each other whether or not the IL was under confinement. As shown in Table 7, the interaction energy between the ionic liquid and the CNT is pretty large under confinement; however, it does not contribute to the transferring energy since the IL–CNT interaction energy does not change significantly when the ionic liquid is transferred from the confined IL phase to the confined CO<sub>2</sub> phase. The IL–CO<sub>2</sub> interaction energies are  $-113.6 \pm 1.2$  kJ/mol under confinement and  $-58.3 \pm 1.5$  kJ/mol without confinement. The IL–CO<sub>2</sub> energy under confinement is about 55.3 kJ/mol stronger than that without confinement, and this strong IL–CO<sub>2</sub> interaction energy is the primary factor that favors the ionic liquid transferring under confinement. The strong IL–CO<sub>2</sub> interaction under confinement is due to the much higher CO<sub>2</sub> density in the (20,20) CNT than the corresponding bulk CO<sub>2</sub> density at the same temperature and pressure. As shown in Figure 14, the confined CO<sub>2</sub> density is much higher than that for the bulk CO<sub>2</sub> and the cation is located closer to the tube wall than the anion. This finding is consistent with the stronger interaction energy between the cation and the CNT than the one between the anion and the CNT, as shown in Table 7.

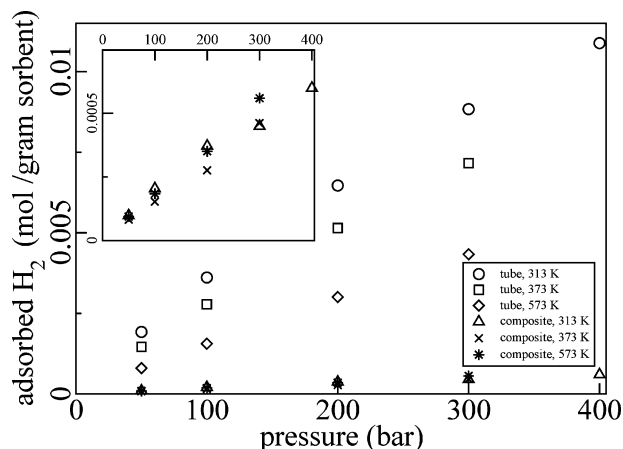
**4.3.3. H<sub>2</sub> Sorption.** The simulated results for H<sub>2</sub> adsorbed in the (20,20) CNT and the composite material are shown in Table 5 and Table 6, while the corresponding adsorption isotherms are shown in Figure 15. The amount of adsorbed H<sub>2</sub> in the (20,20) CNT is about 10–20 times larger than the one in the composite material due to the fact that the void space inside the CNT has been occupied by the ionic liquid in the

**TABLE 7: van der Waals (vdw) and Electrostatic (ELEC) Interaction Energies at 333 K and 48 bar<sup>a</sup>**

|   | C-A (kJ/mol) |         | C-CO <sub>2</sub> (kJ/mol) |          | A-CO <sub>2</sub> (kJ/mol) |          | C-CNT (kJ/mol) | A-CNT (kJ/mol) |
|---|--------------|---------|----------------------------|----------|----------------------------|----------|----------------|----------------|
|   | vdw          | ELEC    | vdw                        | ELEC     | vdw                        | ELEC     | vdw            | vdw            |
| a | −20.0(2)     | 34.8(4) | −29.0(3)                   | −12.2(1) | −40.2(8)                   | −32.2(8) | −65.3(8)       | −17.6(21)      |
| b | −20.3(3)     | 34.3(4) | −15.9(8)                   | −4.3(3)  | −20.1(9)                   | −18.0(8) |                |                |

<sup>a</sup> The results are indicated for cation–anion (C-A) pairs, cation–CO<sub>2</sub> (C-CO<sub>2</sub>), anion–CO<sub>2</sub> (A-CO<sub>2</sub>), cation–carbon nanotube (C-CNT), and anion–carbon nanotube (A-CNT). The system consists of one pair of [hmim][Tf<sub>2</sub>N] plus 463 CO<sub>2</sub> molecules with (a) and without (b) (20,20) CNT confinement. The uncertainty in the last digit is given in parentheses.





**Figure 15.** Amounts of adsorbed  $H_2$  in the (20,20) carbon nanotube and in the composite material consisting of 60 pairs of [hmim][Tf<sub>2</sub>N] molecules and a (20,20) carbon nanotube at 313, 373, and 573 K and at different pressures/axial pressure tensors. The error bars are smaller than the symbols and are not shown. The inset shows the simulated results in the composite material.

composite material and that  $H_2$  solubility in the IL is very small.<sup>25</sup> When compared with the  $H_2$  solubility in the ionic liquid [hmim][Tf<sub>2</sub>N], the amount of  $H_2$  adsorbed in the composite material is typically about 1.1–1.4 times larger than that in the ionic liquid at 313 and 373 K, respectively. However, at a high temperature of 573 K, the amount of adsorbed  $H_2$  in the composite material is about 1.4 times less than that in the ionic liquid. The amount of adsorbed  $H_2$  in the (20,20) CNT is always larger than that in the ionic liquid at all temperatures.

Similar to  $CO_2$ , the Henry's law constants for  $H_2$  sorption in the composite material and in the CNT have been computed using the sorption data shown in Figure 15 at pressures less than 100 bar. The estimated  $H$  values are shown in Table 4. By plotting  $\log(H) \sim 1/T$ , the sorption enthalpy values were calculated to be  $-4.6 \pm 0.2$  kJ/mol in the (20,20) CNT, and  $-0.2 \pm 1.6$  kJ/mol in the composite material. As we have shown before,<sup>25</sup> the absorption enthalpy for  $H_2$  in [hmim][Tf<sub>2</sub>N] is  $3.0 \pm 0.6$  kJ/mol. These enthalpies suggest that, when the temperature is increased, the  $H_2$  sorption amount in the (20,20) CNT decreases, it increases in the bulk [hmim][Tf<sub>2</sub>N], and almost does not change in the composite material as evidenced in Figure 15. The fact that  $H_2$  amount adsorbed in the composite material does not change with temperature appreciably is due to the opposite enthalpy sign of  $H_2$  in the IL and CNT. When the temperature is increased, the confined IL will expand as shown in Figure 1. This expanded volume increases  $H_2$  sorption since  $H_2$  solubility in IL is largely dependent on the IL volume.<sup>25</sup> Simultaneously, higher temperature will lead to lower  $H_2$  sorption in the CNT. Hence, the increased  $H_2$  sorption in the IL and the decreased sorption in the CNT will cancel out. This leads to no significant change of  $H_2$  sorption with temperature in the composite material.

In the composite material, the  $H_2$  solubility increases with pressure and  $H_2$  always behaves as a solute. The IL molecules do not fragment and are not dissolved in  $H_2$  at 313–573 K and 50–400 bar. This is expected since the amount of adsorbed  $H_2$  in the CNT is small and the interaction between the IL and  $H_2$  is very weak.<sup>25</sup> This leads to a transferring energy of one pair of IL from the IL to the  $H_2$  phase under the confinement to be positive and very large and consequently IL molecules will not dissolve in the  $H_2$  under the confinement.

**4.3.4.  $CO_2/H_2$  Selectivity.** We have computed the sorption selectivity of  $CO_2$  over  $H_2$  using the Henry's law constant ratio

**TABLE 8: Sorption Selectivity of  $CO_2$  over  $H_2$  at Different Temperatures in the (20,20) Carbon Nanotube, in the Ionic Liquid [hmim][Tf<sub>2</sub>N], and in the Composite Material Consisting of the Carbon Nanotube and the Ionic Liquid<sup>a</sup>**

| $T$ (K) | $H_{H_2}/H_{CO_2}$ |                |                |
|---------|--------------------|----------------|----------------|
|         | CNT                | IL             | composite      |
| 313.2   | $15.0 \pm 1.0$     | $38.7 \pm 1.6$ | $79.1 \pm 1.8$ |
| 373.2   | $7.8 \pm 0.5$      | $18.7 \pm 2.1$ | $39.8 \pm 6.0$ |
| 573.2   | $2.7 \pm 0.1$      | $2.8 \pm 0.2$  | $6.0 \pm 0.6$  |

<sup>a</sup> Also shown in the table are the corresponding error bars.

of  $H_{CO_2}/H_{H_2}$ . The selectivity values in the (20,20) CNT, the [hmim][Tf<sub>2</sub>N], and the composite material are shown in Table 8. Note that the  $H$  values for  $CO_2$  and  $H_2$  in the ionic liquid are from our previous work.<sup>22,25</sup> At all temperatures, the  $CO_2/H_2$  sorption selectivity exhibits the largest value in the composite material, followed by the ionic liquid and the (20,20) CNT. The only exception from this trend is observed at 573 K where the IL and CNT exhibit  $CO_2/H_2$  selectivity very close to each other. Both  $CO_2$  and  $H_2$  sorptions in the composite material are typically larger than those in the IL. As shown above, the  $CO_2$  sorption in the composite material is about 2–3 times larger than in the IL. In contrast, the  $H_2$  sorption in the composite material is only about 1.1–1.4 times larger than in the IL. This leads to a higher selectivity of  $CO_2/H_2$  in the composite material than in the ionic liquid. When compared with the (20,20) CNT, the composite material exhibits smaller amounts of sorption for both  $CO_2$  and  $H_2$ . The  $H_2$  sorption decreases about 10–20 times in the composite material relative to CNT. However, the  $CO_2$  sorption decreases only 3–4 times. This also leads to a higher  $CO_2/H_2$  selectivity in the composite material than in the (20,20) CNT. Our calculations strongly suggest that the confined [hmim][Tf<sub>2</sub>N] has better performances to separate  $CO_2/H_2$  than either the (20,20) CNT or the IL due to a large sorption selectivity in the composite material.

## 5. Conclusions

Molecular dynamics and Monte Carlo simulations have been used to study the confinement effect on the thermodynamic and transport properties for the [hmim][Tf<sub>2</sub>N] ionic liquid in CNT, and gases sorption in the composite IL–CNT system. In the (20,20) CNT, the cations are first adsorbed close to the tube wall at low loadings. When the loading is increased, the local density peak height for the first layer increases and several layers farther away from the tube wall can occur. Typically, cations are not adsorbed close to the tube center independent of loading. The anions also adsorb first close to the tube wall at low loadings. However, there are significant amounts of anions adsorbed in the tube center at high loadings. The charge density of the IL in the (20,20) CNT is distributed such that it exhibits positive values close to the tube wall and negative ones close to the tube center.

The self-diffusivity coefficient for the [hmim][Tf<sub>2</sub>N] decreases gradually with loading. At 1 bar and 313 K, the [hmim][Tf<sub>2</sub>N] molecule in (20,20) and (9,9) carbon nanotubes diffuses about 300–400 times faster than in the corresponding bulk ionic liquid. At a high temperature of 573 K and at 1 bar, the confined [hmim][Tf<sub>2</sub>N] in the (20,20) CNT diffuses about 14 times faster than in the bulk IL. At 313 K and low loading, the  $H_2$  molecule in the (20,20) CNT diffuses about 4900 times faster than the one in the composite material, and 10 500 times faster than that in the IL. The  $CO_2$  molecule in the (20,20) CNT diffuses about 570 and 9700 times faster than the ones in the composite

material and in the IL, respectively. Overall, the H<sub>2</sub> molecule diffuses about 10 times faster than CO<sub>2</sub> both in the CNT and in the IL. However, in the composite material the H<sub>2</sub> molecule diffuses only about 1.5 times faster than CO<sub>2</sub>.

The amounts of adsorbed CO<sub>2</sub> in the (20,20) CNT are about 3–4 and 6–12 times larger than those in the composite material and the IL, respectively. The CO<sub>2</sub> solubility decreases at elevated temperatures in all three types of materials, i.e., IL, CNT, and the composite IL–CNT system. Very interestingly, the [hmim][Tf<sub>2</sub>N] confined in the (20,20) carbon nanotube behaves very differently than the bulk ionic liquid in terms of dissolving in the CO<sub>2</sub>. In the bulk IL, the CO<sub>2</sub> solubility increases continuously with pressure. The IL always behaves as a solvent and CO<sub>2</sub> as a solute. The IL does not dissolve in the CO<sub>2</sub> phase at temperatures higher than the bulk CO<sub>2</sub> critical temperature. This is due to the very positive unfavorable transferring energy for one pair of IL transferred from the IL phase to the CO<sub>2</sub> phase. In contrast, under confinement the IL dissolves in the CO<sub>2</sub> phase above a pressure of 48 bar and at 333 K. This is due to the negative favorable transferring energy for one pair of IL transferred from the confined IL phase to the confined CO<sub>2</sub> phase. The IL–CO<sub>2</sub> interaction under the confinement is much larger than that without confinement.

For H<sub>2</sub>, the sorption amount in the (20,20) CNT is about 10–20 and 6–25 times larger than those in the composite material and the IL, respectively. By increasing temperature, the H<sub>2</sub> sorption increases in the IL, decreases in the (20,20) CNT, and changes insignificantly in the composite material. The CO<sub>2</sub>/H<sub>2</sub> sorption selectivity exhibits the largest value in the composite material followed by the IL and the CNT. These findings suggest that the composite material will be superior to both the IL and the CNT for CO<sub>2</sub>/H<sub>2</sub> separation applications.

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**Supporting Information Available:** Validation of the FGSF method against the standard Ewald method in the 1-D carbon nanotube. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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